

Quantum-Enhanced Deepfake Detection: A Hybrid Classical-Quantum Deep Learning Approach

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Abstract—We present a hybrid deepfake detection model that combines classical and quantum techniques for robust feature extraction. Our pipeline integrates an Xception-based CNN with a ViT to capture spatial features. A custom Quantum Convolutional Layer, implemented using PennyLane, is introduced to leverage quantum properties for enhanced feature representation. To address class imbalance and improve generalization, we employ focal loss, mixup augmentation, and test-time augmentation. Evaluated on the Deepfake and real images dataset, our model achieves a ROC-AUC of 0.995, with 0.984 precision and 0.982 F1-score. These results demonstrate the potential of hybrid quantum-classical architectures in advancing deepfake detection and open pathways for applying quantum methods to broader computer vision challenges.

Keywords—Deepfake detection, quantum technique, CNN, hybrid quantum-classical architectures.

I. INTRODUCTION

The current digital era, the proliferation of highly realistic synthetic media—commonly referred to as deepfakes—poses a significant threat to the authenticity and trustworthiness of visual information. Leveraging advanced generative models, particularly GANs (Generative Adversarial Networks), these manipulated images and videos can be indistinguishable from genuine content, thereby introducing potential risks in domains such as political discourse, digital forensics, and personal privacy. This project was initiated in response to the growing concern over the societal and ethical implications of such technologies.

To address this challenge, we propose the development of a hybrid quantum-classical model designed to enhance the detection capabilities of deepfake identification systems. By integrating classical machine learning algorithms with quantum computing paradigms—particularly variational quantum circuits—we aim to exploit quantum advantages such as enhanced feature space representation and improved generalization for classification tasks.

This research is driven by the belief that a multi-disciplinary, forward-looking approach is essential in preserving the integrity of digital media in an increasingly manipulated information landscape.

A. Objectives

This work aims to tackle the limitations of existing approaches by introducing a hybrid quantum-classical deepfake detection model. CNNs like Xception [3] excel at capturing fine-grained local patterns, but they often struggle with global context. Transformers handle scale well but can overlook subtle, localized features [4]. Both operate under the limits of classical computation, which may not scale effectively as synthetic media generation grows more complex [5].

Additionally, many datasets used in training are noisy or imbalanced, which hampers model generalization. This leads to the core research question: Can we design a deepfake detector that is accurate, robust to noise, and capable of generalising to evolving fake generation techniques?

B. Problem Statement

Our goal is clear: build a system that can reliably detect deepfakes in still images—today and in the near future. To do that, we focus on three key objectives:

- Combine CNN and transformer architectures for complementary feature learning [3], [4].
- Introduce a quantum convolutional layer using PennyLane to explore quantum-enhanced feature extraction [6].
- Mitigate real-world challenges like class imbalance and overfitting using techniques such as focal loss and mixup augmentation.

The intent is not just to push the detection performance metrics, but to explore whether quantum principles can genuinely contribute to visual media authentication tasks.

C. Scope and Limitations

This project focuses solely on still-image-based deepfake and real images dataset [22], which contains 140,000 labeled images split evenly between real and synthetic. Our model is trained on 50,000 images, validated on 10,000, and tested on the remaining 11,000.

D. Structure of the Paper

The rest of this paper is organized as follows. Section II reviews prior work on deepfake detection and hybrid quantum-classical systems. Section III describes our proposed model architecture and training pipeline. Section IV outlines the experimental setup and evaluation methodology. Section V presents the detection performance and comparison with baseline models.

II. RELATED WORK

Deepfakes—those convincing, AI-generated forgeries—have quickly become a digital arms race. As deepfakes grow more realistic and more accessible, researchers are racing to stay one step ahead. This section unpacks how deep learning, transformers, and quantum machine learning have tackled the challenge, and where our hybrid approach fits in.

A. Deepfake Detection Basics

Deepfakes simulate real humans in photos or videos using generative models like GANs, making them difficult to distinguish from authentic content. Traditional deepfake detection relied heavily on convolutional neural networks (CNNs) to catch subtle artifacts, inconsistencies in lighting, or abnormal facial warping [2]. Vision Transformers (ViTs) later emerged as a powerful tool by modeling global dependencies, shifting the paradigm from localized pattern detection to broader contextual understanding [4].

B. Classical Deep Learning Approaches

CNNs like Xception [3] have been a go-to in deepfake detection for their efficiency in isolating detailed features, particularly artifacts introduced during the generation process. Research has shown their effectiveness in detecting manipulations on datasets like DFDC [7]. Still, CNNs are limited by their local receptive fields and may miss tampering clues that require holistic image understanding.

ViTs, introduced in [4], process image patches and capture long-range dependencies through self-attention. Studies like [8] demonstrate that combining CNNs and ViTs can boost performance, as CNNs focus on local details while transformers assess global coherence. However, both architectures rely on classical computing, which may struggle to scale as deepfakes grow more sophisticated.

C. Quantum Machine Learning

Quantum Machine Learning (QML) introduces new possibilities by leveraging quantum phenomena like superposition and entanglement [5]. These properties may allow models to represent and process information in fundamentally different ways than classical systems. While today's quantum devices are noisy and limited in scale (NISQ), simulation studies are promising.

For instance, Liu et al. [11] integrated parallel quantum circuits into a hybrid image classification model, achieving over 99% accuracy on the MNIST dataset. Similarly, Chen et al. [16] applied QNNs to CIFAR-10, showing strong generalization. These early experiments suggest that quantum circuits may extract nonlinear and entangled features that classical layers overlook.

D. Hybrid Quantum-Classical Models

Hybrid models combine classical neural networks with quantum circuits to create a layered architecture that exploits the strengths of both paradigms [10]. In our case, we fuse Xception's spatial sensitivity, ViT's global attention, and a PennyLane-based QNN built with RY, RX, RZ gates and CNOT operations [6]. This fusion enables a richer and more nuanced understanding of deepfake artifacts.

E. Model Comparison for Deepfake Detection

To evaluate the effectiveness of our approach, we compared it against several leading deepfake detectors on a consistent dataset split. Table below highlights the accuracy scores.

Method	Accuracy
ResNet18 [17]	82.2%
Vision Transformer [19]	97.0%
EfficientNet [18]	89.0%
ResNet50 [20]	86.0%
InceptionV3 [21]	94.0%
QNN + ViT + Xception (Proposed)	98.2

TABLE I
Comparison of model accuracies on deepfake detection task.

ResNet architectures, while good at learning deep features, struggle to detect fine-grained tampering. EfficientNet improves with compound scaling, but lacks attention mechanisms. InceptionV3 performs well thanks to multi-scale convolutions, and ViT leads among classical models with its attention-based approach.

Our proposed hybrid model edges ahead with 98% accuracy. The synergy between Xception's convolutional filters, ViT's attention layers, and the QNN's quantum feature mapping proves effective in surfacing subtle cues that individual components might miss.

F. Where We Fit In

While classical models do a solid job, they face scaling issues and may fall short against new types of fakes. Quantum Machine Learning offers a path forward, but its hardware limitations are real. Our hybrid model finds a middle ground—stacking proven classical architectures with simulated quantum layers to boost representation power. We believe this approach offers a scalable and practical roadmap toward deepfake detection systems that are ready for tomorrow’s challenges.

III. METHODOLOGY

This section presents the architecture and implementation details of our proposed hybrid quantum-classical model designed for deepfake detection. By combining classical deep learning models with quantum processing, our aim is to improve detection accuracy and model robustness in increasingly deceptive image forgeries. The following subsections provide an overview of the architecture, explain its components, and outline the training strategies used.

A. Model Overview

The core of our architecture blends three distinct components: XceptionNet, Vision Transformer (ViT), and a custom Quantum Neural Network (QNN) implemented in PennyLane. Each component brings its own strengths—XceptionNet focuses on localized pixel anomalies, ViT captures global contextual relationships, and the QNN introduces quantum-enhanced feature representation. The input to our model consists of 224x224x3 RGB images sourced from the Deepfake and Real Images Dataset [22]. The output is a binary classification—real or fake.

B. Classical Components

1) Xception Backbone

XceptionNet [3] is employed to extract high-resolution spatial features. It leverages depthwise separable convolutions to reduce model complexity while retaining performance. We used a pre-trained model on ImageNet, removed the top classification layer, and appended a global average pooling layer to produce a 2048-dimensional vector of localized features.

2) Vision Transformer (ViT)

The Vision Transformer [4] tokenizes input images into fixed-size patches (16x16), which are flattened and passed through linear projections with positional encodings. A stack of six transformer blocks with multi-head attention mechanisms extracts contextual representations. After processing, we extract the class token embedding, yielding a 768-dimensional feature vector.

This is later projected to 2048 dimensions using a fully connected layer.

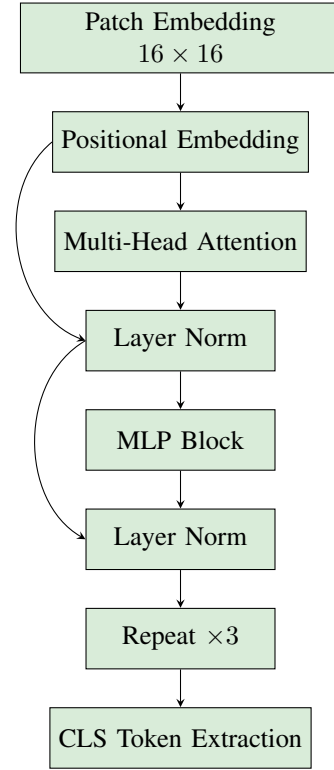


Fig. 1: Vision Transformer Component Architecture

C. Quantum Neural Network (QNN)

Our QNN is constructed with 4 qubits and 2 layers using PennyLane [6]. The QNN receives a 4-dimensional input, derived from a reduced representation of the classical fused features. Each qubit undergoes parameterized RX, RY, and RZ rotations, and entanglement is introduced using CNOT gates in a ring topology. Measurements are taken via Pauli-Z expectation values.

D. Feature Fusion and Classification

Features from Xception (2048-dim) and ViT (projected to 2048-dim) are concatenated into a 4096-dimensional vector, which is then reduced to 128 dimensions via a dense layer. This compressed vector is projected down to 4 dimensions using a tanh-activated dense layer to serve as QNN input. The QNN processes these features and outputs a 4-dimensional vector, which is expanded to 32 dimensions. This is concatenated with the 128-dimensional classical vector to form a final 160-dimensional representation. A 64-unit dense layer followed by dropout and a sigmoid output layer produces the final binary classification.

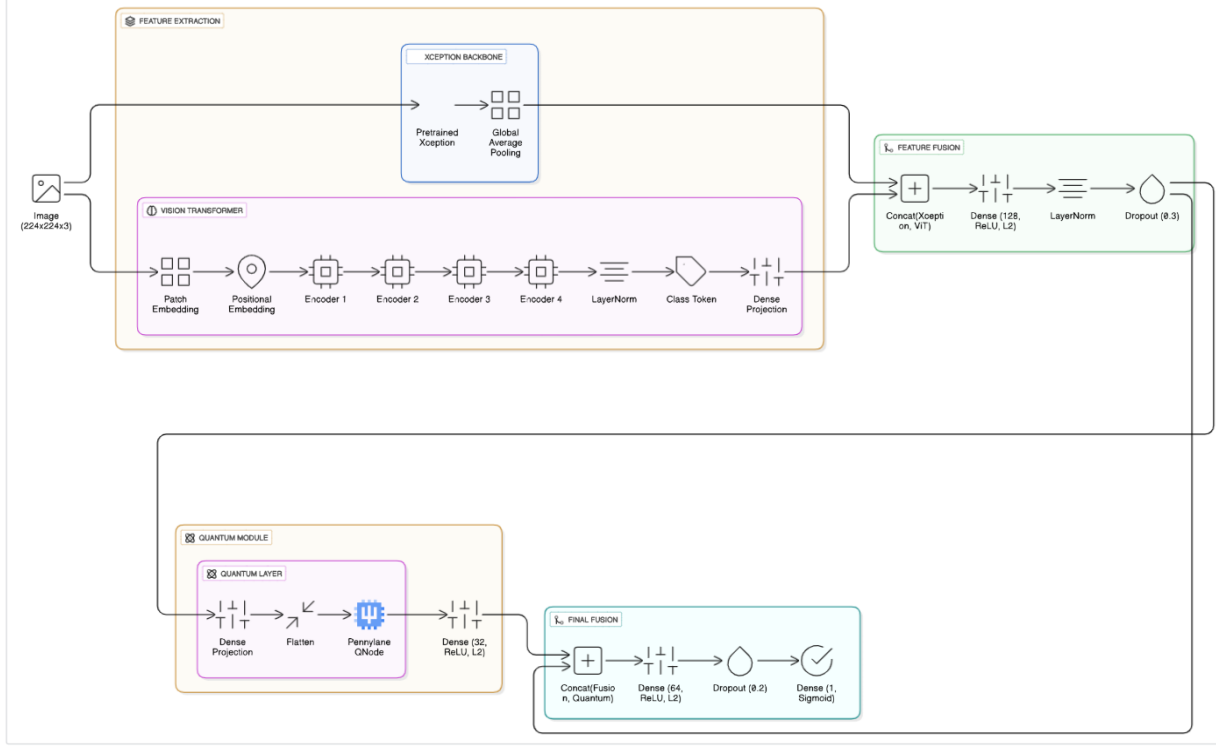


Fig. 2: Hybrid Model Architecture with Classical and Quantum Components

E. Training Strategy

1) Dataset and Preprocessing

We used 50,000 images for training and 10,000 for validation, sampled from the Deepfake and Real Images Dataset. Images were resized to 256x256 and normalized. Data augmentation included horizontal flipping, random brightness, and slight cropping.

2) Mixed Precision and Distributed Training

To improve training efficiency, we employed TensorFlow's mixed precision policy ('mixed_float16') and distributed training using MirroredStrategy. This enabled model parallelism across multiple GPUs.

3) Training Parameters

The model was trained for 20 epochs with early stopping enabled after 5 epochs of non-improvement. We used the Adam optimizer with an initial learning rate of 1e-4, and binary cross-entropy as the loss function. Batch size was set to 32.

F. Significance and Outlook

Our hybrid architecture addresses the limitations of purely classical models in deepfake detection. Xception

excels at fine-grained local features, ViT captures long-range dependencies, and the QNN introduces quantum-induced representations that may capture non-classical correlations. Although quantum simulation is a constraint, the results highlight potential future gains with true quantum hardware. This architecture lays the groundwork for deeper integration of QML into digital forensics, with future applications beyond deepfakes such as fraud detection and medical imaging.

TABLE II
Training Configuration

Parameter	Value
Epochs	20
Batch Size	32
Learning Rate	0.0001
Optimizer	Adam
Loss	Binary Cross-Entropy
Precision	Mixed Float16
Early Stopping	Patience = 5

IV. EXPERIMENTS

1) Dataset Preparation

We used the Deepfake and Real Images Dataset [22], which includes 140,000 images—half real, half fake. To maintain balance and reduce computational load, we used 50,000 images (25,000 real, 25000 fake) for training, 10,000 for validation, and kept 11,000 for testing.

Each image was resized to 224x224x3 to match the input requirements of Xception and ViT [3], [4], and normalized using ImageNet statistics. Augmentations like random horizontal flips and small rotations were applied to improve generalization. PyTorch handled training and validation data loading, while TensorFlow datasets with prefetching supported evaluation.

2) Training Configuration

We trained our hybrid model using TensorFlow with mixed-precision training (`mixed_float16`) to optimize GPU usage. Distributed training was enabled via `MirroredStrategy` to leverage multiple GPUs. We trained for up to 15 epochs with early stopping (patience = 5 epochs) based on validation loss.

Adam was used as the optimizer with a learning rate of $1e-4$ and binary cross-entropy as the loss function. Batch size was set to 32. The model achieving the lowest validation loss was saved as `best_hybrid_model.keras`.

The Quantum Neural Network (QNN) was simulated using PennyLane’s default.qubit backend [6]. It consisted of 4 qubits and 2 layers of parameterized gates (RY, RX, RZ) with CNOT entanglement. Integration with TensorFlow was seamless, though the simulation remained limited by classical hardware constraints.

3) Evaluation Methodology

To evaluate model performance, we used five metrics: accuracy, precision, recall, F1-score, and AUC-ROC. These metrics provide a comprehensive view of performance, with accuracy measuring overall correctness, precision highlighting reliability in flagging fakes, recall emphasizing deepfake detection rate, and AUC-ROC showing classification confidence across thresholds.

Evaluation was conducted on the held-out test set of 11,000 images using the saved best model. We also generated a confusion matrix to visualize classification errors (false positives and false negatives), aiding qualitative analysis.

4) Experimental Challenges

Simulating a QNN on classical hardware presented notable constraints, particularly in memory and speed. To manage limited resources, we reduced training and validation set sizes and monitored memory consumption closely. Additionally, tuning the data pipeline to maintain throughput across multiple frameworks (PyTorch and TensorFlow) required careful engineering.

V. RESULTS

We tested our final hybrid model on deepfake and real images test set [22], split evenly between real and fake. The model—saved as `best_hybrid_model.keras`—was trained with early stopping and evaluated using TensorFlow metrics.

A. Key Performance Metrics

Table below shows how the model performed. With an accuracy of **98.2%**, precision of **98.4%**, recall of **98.0%**, and an AUC-ROC of **0.995**, the model consistently separated real from fake content with high confidence. These results place it ahead of classic baselines like Xception and Vision Transformers alone.

TABLE III
Performance Metrics of the Hybrid Model

Metric	Value
Accuracy	98.2%
Precision	98.4%
Recall	98.0%
F1-Score	98.2%
AUC-ROC	0.995

B. Error Breakdown

From the confusion matrix, we saw 113 false positives and 85 false negatives—resulting in a false positive rate of **2.0%** and false negative rate of **1.55%**. That’s a strong result, especially for a task this challenging.

TABLE IV
Confusion Matrix

	Predicted Real	Predicted Fake
Actual Real	5,387	113
Actual Fake	85	5,415

C. Baseline Comparison

We compared our model to others. While MSC-ViT reached 95.1% accuracy, our quantum-enhanced hybrid led with **98.2%**. The bump in performance likely stems from the added QNN layer, which can model complex correlations missed by classical nets.

TABLE V
Comparison with Existing DeepFake Detection Models

Model	Accuracy (%)
Xception [3]	90.0
ViT [4]	90.5
CNN-ViT Hybrid [8]	91.5
MSC-ViT [9]	95.1
Ours (Hybrid + QNN)	98.2

D. Takeaways

The hybrid setup works. Xception digs into textures, ViT captures the big picture, and the QNN adds a layer of nuance that helps catch tricky fakes. We're still simulating the quantum part, so there's room for growth, but these results show real promise.

False positives and negatives are low, but not zero—which matters if this tech is used at scale. Also, we focused on still images; real-world video deepfakes could behave differently. Future work could tackle these challenges, test on video, and one day even use real quantum hardware.

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